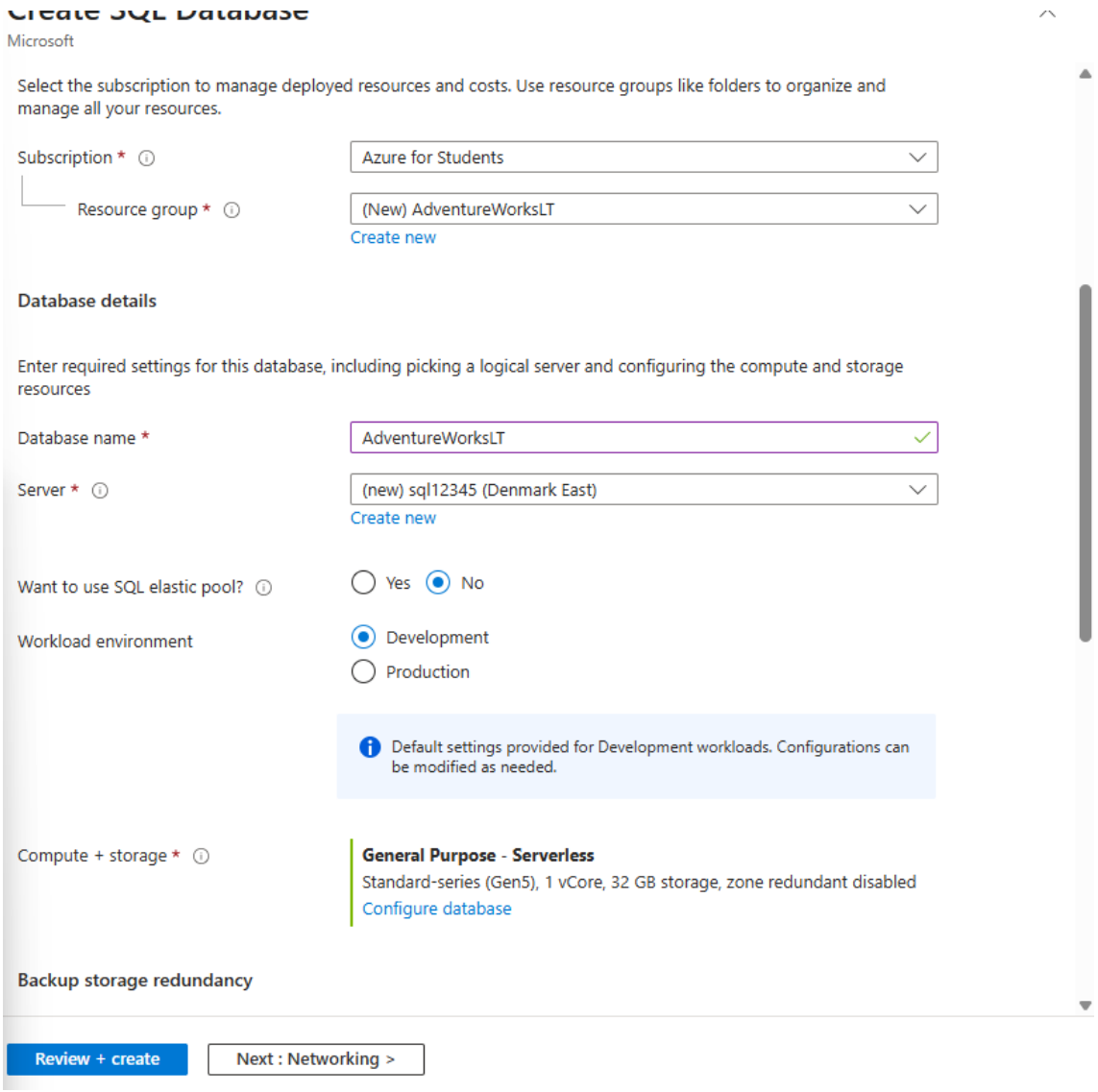
Provision an Azure SQL Database



Configure your development environment

```
PowerShell 7.6.6
PS C:\Users\AlumnoM_AI> gh --version
gh version 2.102.0 (2026-09-30)
https://github.com/cli/cli/releases/tag/v2.102.0
PS C:\Users\AlumnoM_AI> dotnet --version
10.0.400
PS C:\Users\AlumnoM_AI> git --version
git version 2.55.0.windows.3
PS C:\Users\AlumnoM_AI>
```

```
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC> gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! One-time code (5222-39C6) copied to clipboard
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as isaac-sanchez-4
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC>
```

```
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC>  dotnet new install Microsoft.Build.Sql.Templates

Esto es .NET 10.0.
---------------------
Versión del SDK: 10.0.400

Telemetría
---------
Las herramientas de .NET recopilan datos de uso para ayudarnos a mejorar su experiencia. Microsoft los recopila y los comparte con la comunidad. Pued
 DOTNET_CLI_TELEMETRY_OPTOUT en "1" o "true" mediante su shell favorito.

Lea más sobre la telemetría de las herramientas de la CLI de .NET: https://aka.ms/dotnet-cli-telemetry

---------------
Instalar un certificado de desarrollo HTTPS de ASP.NET Core.
Para confiar en el certificado, ejecute "dotnet dev-certs https --trust"
Obtenga información sobre HTTPS: https://aka.ms/dotnet-https

---------------
Escribir su primera aplicación: https://aka.ms/dotnet-hello-world
Descubra las novedades: https://aka.ms/dotnet-whats-new
Explore la documentación: https://aka.ms/dotnet-docs
Notificar problemas y encontrar el código fuente en GitHub: https://github.com/dotnet/core
Use "dotnet --help" para ver los comandos disponibles o visite: https://aka.ms/dotnet-cli
--------------------------------------------------------------------------------
Se instalarán los siguientes paquetes de plantillas:
   Microsoft.Build.Sql.Templates
```

Create a SQL database Project

```
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC>  cd AdventureWorksDB
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC\AdventureWorksDB>  dotnet n
ew sqlproj -tp SqlAzureV12
La plantilla "SQL Server Database Project" se creó correctamente.

PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC\AdventureWorksDB>
```

Welcome | InventoryLog.sql | uspLogInventoryChange.sql

AdventureWorksDB > Tables > InventoryLog.sql

Connect to MSSQL

```sql
CREATE TABLE [dbo].[InventoryLog] (
    [LogID] INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
    [ProductID] INT NOT NULL,
    [ChangeDate] DATETIME2 NOT NULL DEFAULT GETDATE(),
    [QuantityChange] INT NOT NULL,
    [ChangeType] NVARCHAR(20) NOT NULL
);
```

Welcome | InventoryLog.sql | uspLogInventoryChange.sql

AdventureWorksDB > Tables > uspLogInventoryChange.sql

Connect to MSSQL

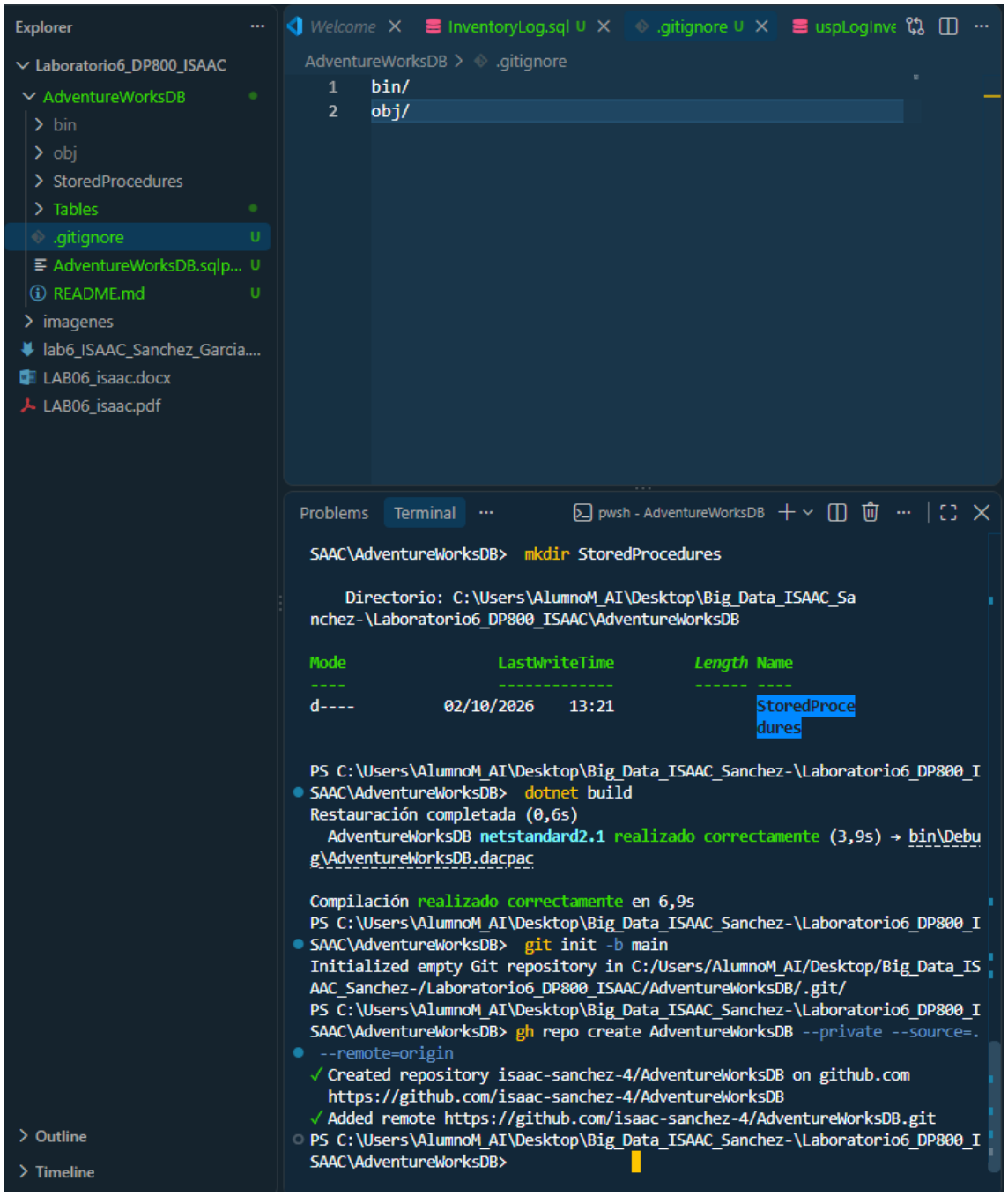
```sql
CREATE PROCEDURE [dbo].[uspLogInventoryChange]
    @ProductID INT,
    @QuantityChange INT,
    @ChangeType NVARCHAR(20)
AS
BEGIN
    INSERT INTO [dbo].[InventoryLog] (ProductID, QuantityChange, ChangeType)
    VALUES (@ProductID, @QuantityChange, @ChangeType);
END;
```

Build the project locally

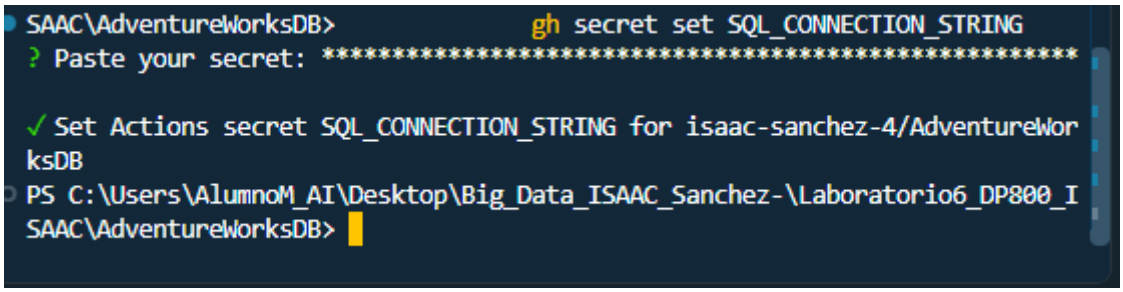
```
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC\AdventureWorksDB>  dotnet b
uild
Restauración completada (0,6s)
  AdventureWorksDB netstandard2.1 realizado correctamente (3,9s) → bin\Debug\AdventureWorksDB.dacpac

Compilación realizado correctamente en 6,9s
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_ISAAC\AdventureWorksDB>
```

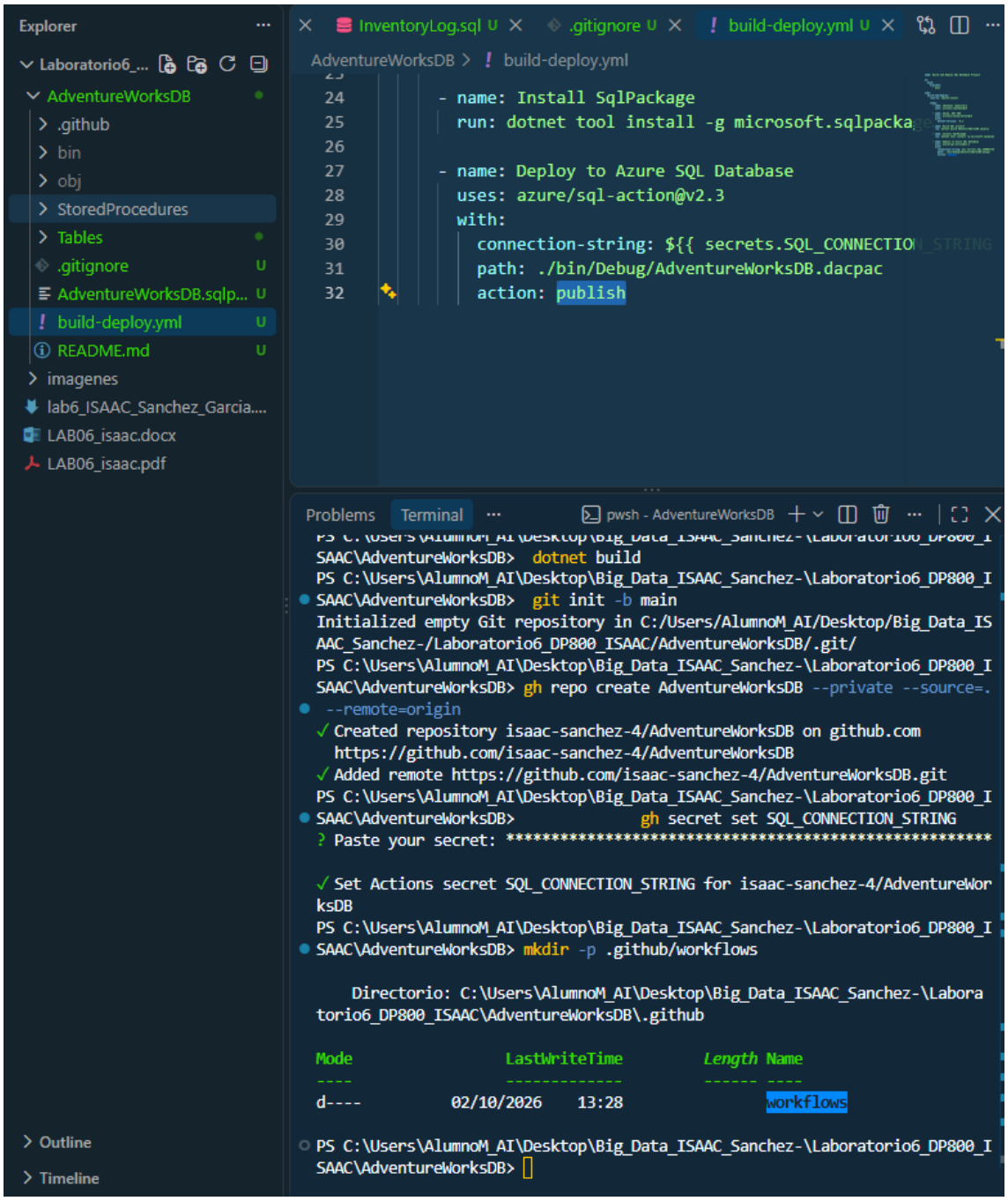
Initialize Git and create a GitHub repository



Configure the GitHub secrets for deployment

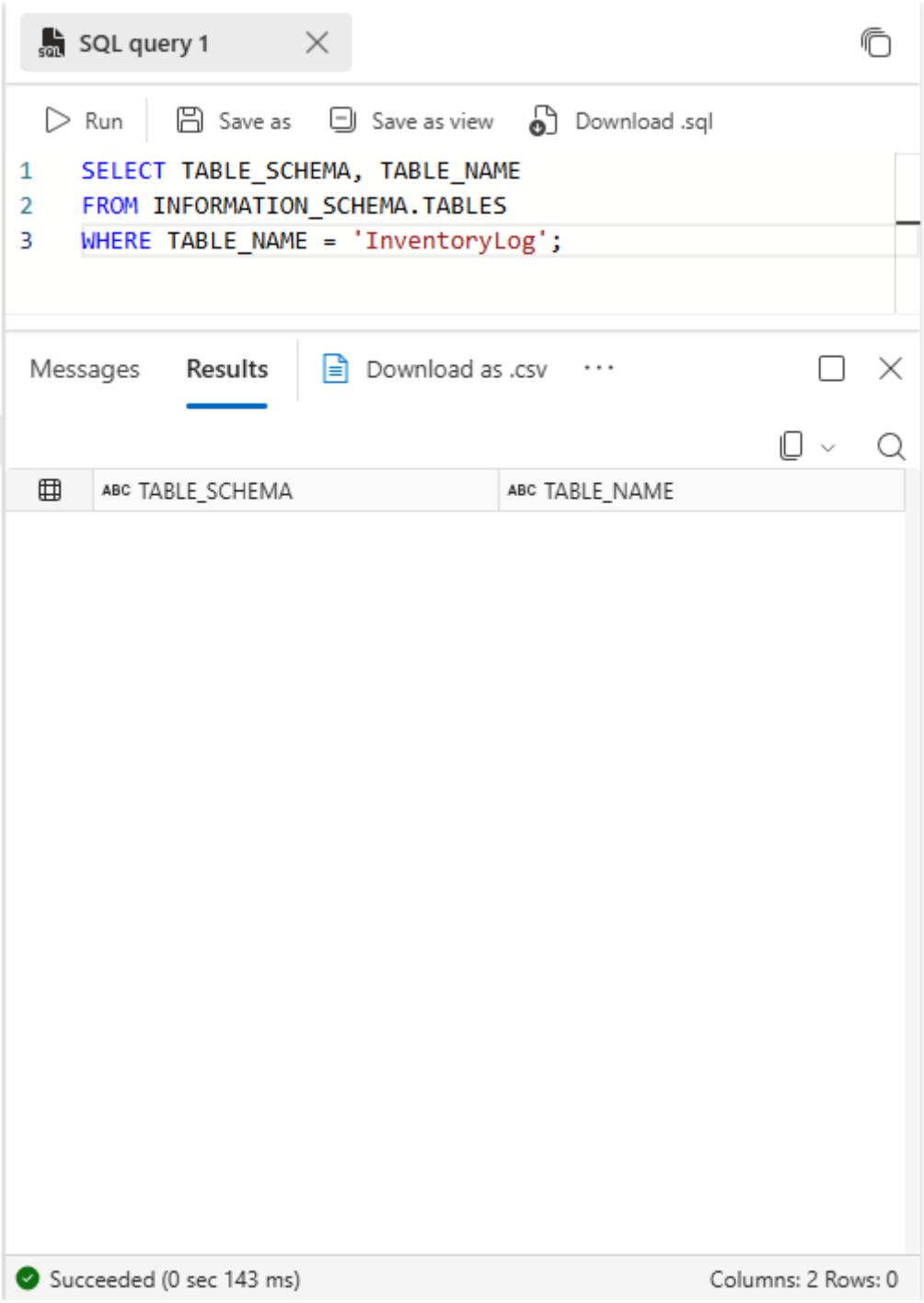


Create the GitHub Actions workflow

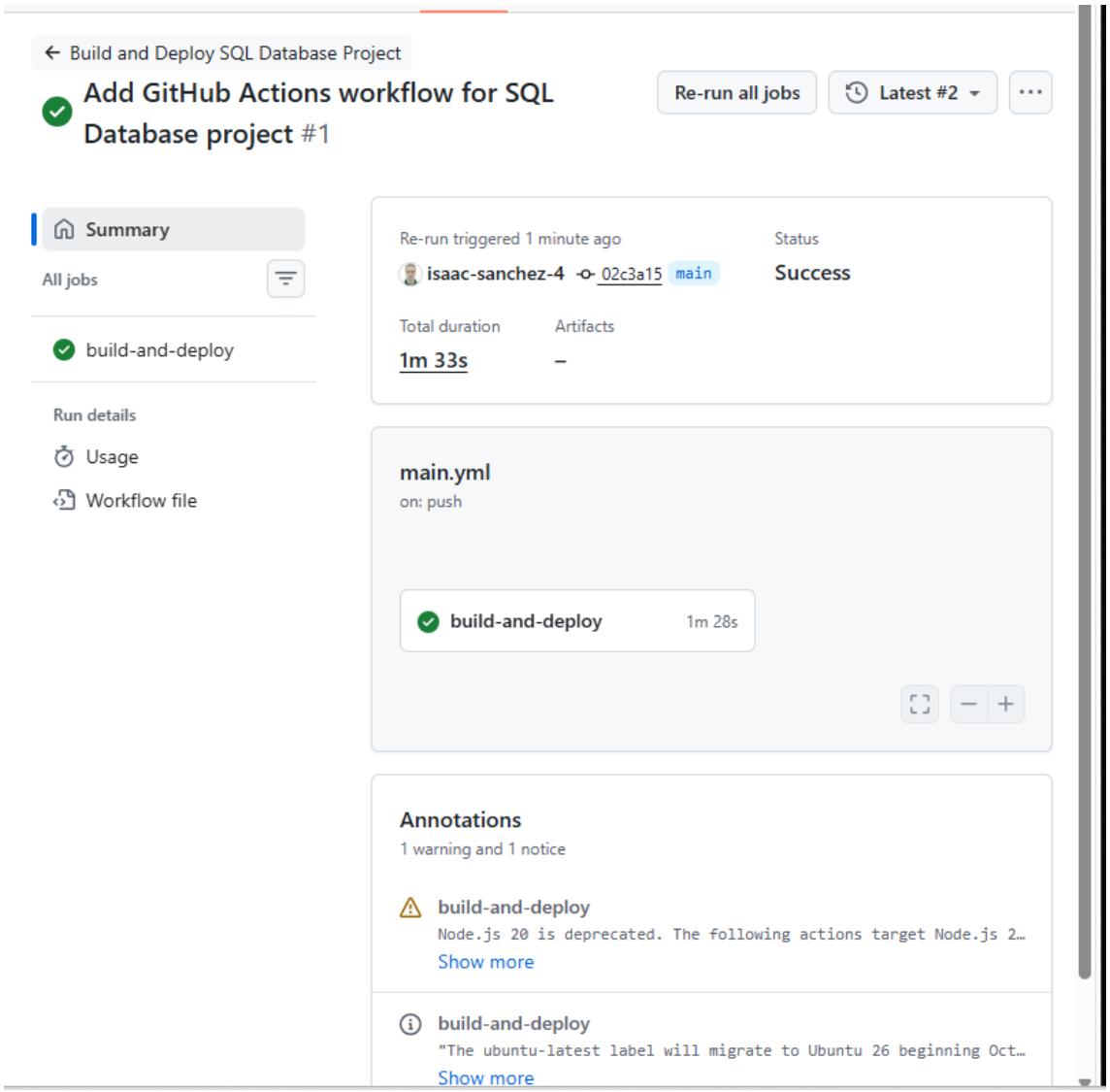


Push and verify the first deployment

```
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_I
SAAC\AdventureWorksDB> git add -A
warning: in the working copy of 'README.md', LF will be replaced by CRLF th
e next time Git touches it
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_I
SAAC\AdventureWorksDB> git commit -m "Initial SQL project with InventoryLog
 table and CI/CD pipeline"
[main (root-commit) 45a7275] Initial SQL project with InventoryLog table an
d CI/CD pipeline
 6 files changed, 89 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 AdventureWorksDB.sqlproj
 create mode 100644 README.md
 create mode 100644 Tables/InventoryLog.sql
 create mode 100644 Tables/uspLogInventoryChange.sql
 create mode 100644 build-deploy.yml
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_I
SAAC\AdventureWorksDB> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 2.00 KiB | 2.00 MiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/isaac-sanchez-4/AdventureWorksDB.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\AlumnoM_AI\Desktop\Big_Data_ISAAC_Sanchez-\Laboratorio6_DP800_I
SAAC\AdventureWorksDB>
```



Push a schema change


Build and Deploy SQL Database Project
Add GitHub Actions workflow for SQL Database project #1
Re-run all jobs
Latest #2
Summary
All jobs
build-and-deploy
Run details
Usage
Workflow file
Re-run triggered 1 minute ago
isaac-sanchez-4
02c3a15
main
Status
Success
Total duration
1m 33s
Artifacts
–
main.yml
on: push
build-and-deploy
1m 28s
Annotations
1 warning and 1 notice
build-and-deploy
Node.js 20 is deprecated. The following actions target Node.js 2…
Show more
build-and-deploy
"The ubuntu-latest label will migrate to Ubuntu 26 beginning Oct…
Show more

Run | Save as | Save as view | Download .sql

```sql
SELECT TABLE_SCHEMA, TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_NAME = 'InventoryLog';
```

Messages | Results | Download as .csv

	TABLE_SCHEMA	TABLE_NAME
1	dbo	InventoryLog

SQL query 1

Run | Save as | Save as view | Download .sql

```sql
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'InventoryLog'
ORDER BY ORDINAL_POSITION;
```

Messages | Results | Download as .csv

	COLUMN_NAME	DATA_TYPE	IS_NULLABLE
1	LogID	int	NO
2	ProductID	int	NO
3	ChangeDate	datetime2	NO
4	QuantityChange	int	NO
5	ChangeType	nvarchar	NO